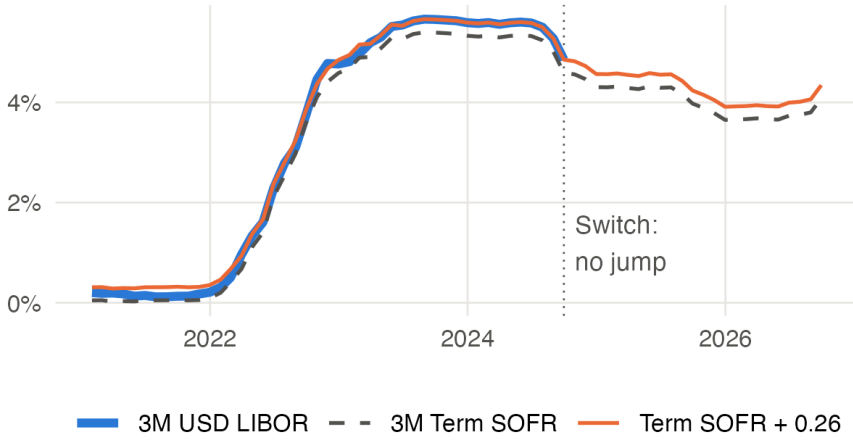
The data

Asset class	Index (total return)	Ccy
US equity	MSCI USA Net	USD
Euro area equity	MSCI EMU Net	EUR
Dev. ex US ex euro area equity	Built from World ex USA and EMU	USD
Emerging markets equity	MSCI EM Net	USD
Euro government bonds	Bloomberg Euro Agg. Treasury	EUR
US Treasuries	Bloomberg US Treasury	USD
Euro IG corporate bonds	Bloomberg Euro Agg. Corporate	EUR

Monthly Bloomberg data from Jan 1999 to Sep 2026. Cash: 3M Euribor and 3M USD rate.

USD cash rate: LIBOR ended in Sep 2024



Continued with Term SOFR + 0.26 pp (ISDA fallback)

1 Developed ex US ex euro area

MSCI World ex USA minus the euro area, at last month's market-cap weight w:

$$r_t^{\text{rest}} = \frac{r_t^{\text{World ex US}} - w_{t-1}\, r_t^{\text{EMU}}}{1 - w_{t-1}}$$

w = euro area share of World ex USA market cap (26 - 42% over the sample)

2 Euro excess return (currency policy)

$$r_{i,t}^{e} = (1 + r_{i,t}^{\text{USD}})(1 + e_t) - 1 - h_i\, x_t - i_{t-1}^{\text{EUR}}/12$$

h = ½ for USD equities, 1 for US Treasuries, 0 for euro assets (25.5% USD short)

Performance Metrics

Asset class	Mean	Vol.	Sharpe	Worst month	Max DD
US eq.	6.8%	14.7%	0.46	-14.0% (Oct 2008)	-66.4%
Euro eq.	4.9%	17.2%	0.28	-17.8% (Sep 2002)	-63.3%
Dev. eq.	4.6%	13.5%	0.34	-16.2% (Oct 2008)	-54.1%
EM eq.	8.2%	19.5%	0.42	-25.3% (Oct 2008)	-62.1%
Euro govt	1.3%	4.5%	0.28	-5.2% (Aug 2022)	-24.7%
US Tsy	0.7%	4.6%	0.16	-4.6% (Jul 2003)	-27.0%
Euro corp	1.6%	4.1%	0.40	-6.8% (Mar 2020)	-16.6%
Long USD*	1.3%	9.3%	0.14	-9.3% (Dec 2008)	-46.3%

Euro excess returns under the currency policy, annualized, Jan 1999 - Sep 2026.

* Currency excess return x: long USD financed in EUR.

Equities earn 5x bonds

- at ~4× the risk

Losses cluster in crises

- Worst months in Sep 2002 and Oct 2008

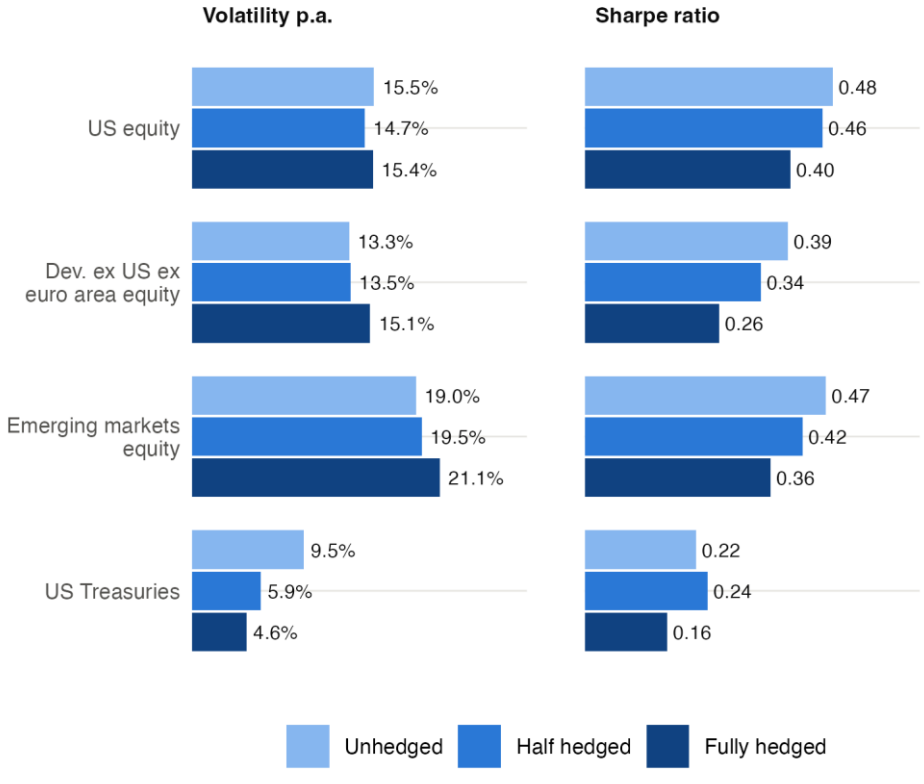
Hedge the bonds

- US Treasury vol. falls from 9.5% to 4.6%.
- The half hedge raises vol. for developed and EM equity.

Means vs. the long run

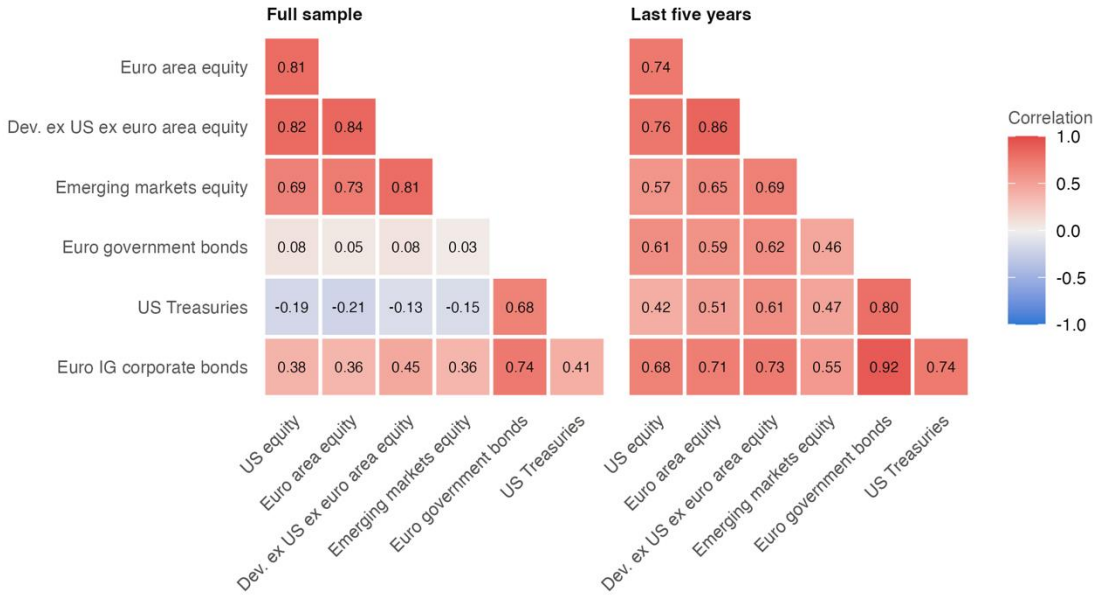
- CIs too wide to use as expectations (US equity: 1.3-12.2%).
- Falling-yield bond gains unlikely to repeat

Effect of the currency hedge on the USD assets



Current policy: half hedged for equities, fully hedged for US Treasuries

Correlation Structure



Volatility p.a.	Full	Last 5y	EWMA
US equity	**14.7%**	14.9%	13.7%
Euro area eq.	**17.2%**	14.1%	13.6%
Dev. ex US eq.	**13.5%**	12.5%	12.0%
EM equity	**19.5%**	16.5%	17.3%
Euro govt	**4.5%**	6.5%	5.4%
US Treasuries	**4.6%**	5.6%	4.9%
Euro corp	**4.1%**	5.2%	4.3%
SAA (ex ante)	**7.23%**	**8.28%**	**7.38%**

Annualized; SAA = ex-ante portfolio volatility, limit 9%.
Full sample is our baseline.

Would not extrapolate

- Stock-bond corr.: 0.05 to 0.59 (euro), -0.19 to 0.42 (US)
- Low bond vol.: euro govt 4.5% vs. 6.5% in the last 5y
- Govt-corp corr.: 0.74 vs. 0.92, may fall in a credit event

Our choice

- Full-sample Σ as the long-run baseline.
- Since it spans several regimes, including crises.
- EWMA as robustness check for Π and the 9% limit.

SAA Equilibrium returns

1 Assume the SAA is optimal

$$\Pi = A \Sigma w_{\text{SAA}}$$

2 Calibrate A to a 2.7% SAA premium

$$A = \frac{0.027}{\sigma^2_{\text{SAA}}} = \frac{0.027}{0.0723^2} = 5.17$$

3 Implied Sharpe ratio

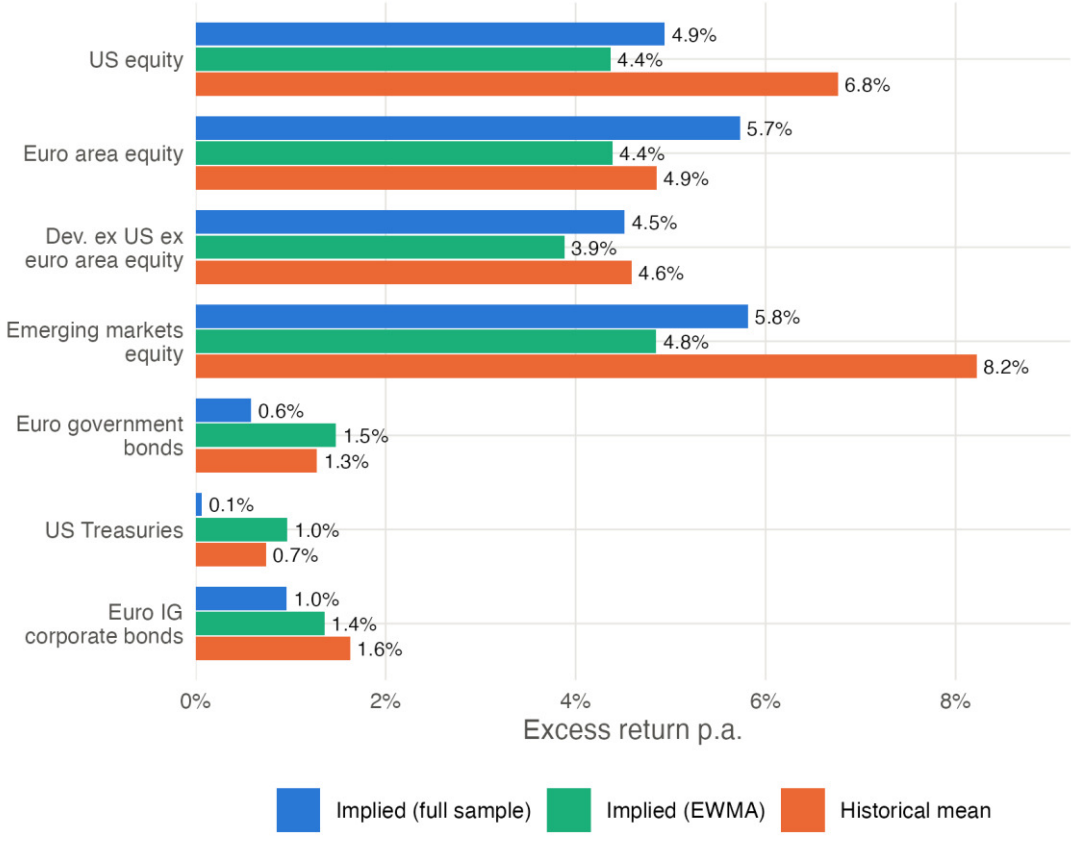
$$\frac{\Pi_i}{\sigma_i} = \underbrace{\frac{0.027}{\sigma_{\text{SAA}}}}_{0.37} \times \rho_{i,\text{SAA}}$$

0.30-0.34

implied Sharpe ratio of the equity classes.

Why equities?

Equities are 45% of the weight but 87% of the risk, so they move with the SAA (ρ ≈ 0.9) and earn the most. Treasuries barely move with it (ρ = 0.04), so they earn ≈ 0.



Implied excess returns Π vs. historical means, % p.a. (A recalibrated for EWMA)

The Four views

1. **Regress** next-12-month excess returns on today's predictor.
2. **View v** = fitted value at Sep 2026 (Newey–West SEs).
3. **ω** = variance of the fitted mean, scaled up after a poor out-of-sample record.

View, Oct 2026 – Sep 2027	Type	Predictor today	View v	Equilibrium	√ω	Prior √(τp'Σp)
US equity	Absolute	CAPE 40.6, D/P 1.07%	1.68%	4.94%	3.00%	3.28%
Euro government bonds	Absolute	CP factor 0.30%	1.50%	0.58%	0.93%	1.01%
US Treasuries	Absolute	Term spread 1.18%	0.92%	0.06%	0.82%	1.04%
Euro corp. minus euro govt bonds	Relative	OAS 0.91%	-0.47%	0.37%	0.54%	0.70%

Black–Litterman form: $P\mu = v + \varepsilon,\ \varepsilon \sim N(0, \Omega)$

$$P = \begin{array}{l} \\ \text{1 US equity} \\ \text{2 Euro govt} \\ \text{3 US Treasuries} \\ \text{4 Corp - govt} \end{array} \begin{array}{c} \begin{array}{ccccccc} \text{US} & \text{EA} & \text{Dev} & \text{EM} & \text{Govt} & \text{UST} & \text{Corp} \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{pmatrix} \end{array} \quad v = \begin{pmatrix} 1.68\% \\ 1.50\% \\ 0.92\% \\ -0.47\% \end{pmatrix} \quad \omega = \begin{pmatrix} 0.00090 \\ 0.00009 \\ 0.00007 \\ 0.00003 \end{pmatrix}$$

Ω = diag(ω). Absolute views have a single 1; the relative view is +1/-1, so its row sums to zero. All views are euro excess returns under the currency policy.

Weak forecasts, larger ω

- 4 of 5 slopes significant at 5%
- Only the term and credit spreads beat the historical mean out of sample.

Views as certain as the prior

- √ω ≈ 0.78-0.92 × the prior uncertainty for every view
- so views and equilibrium get similar weight in task 5.

View 1: Record US equity valuations

1 Idea

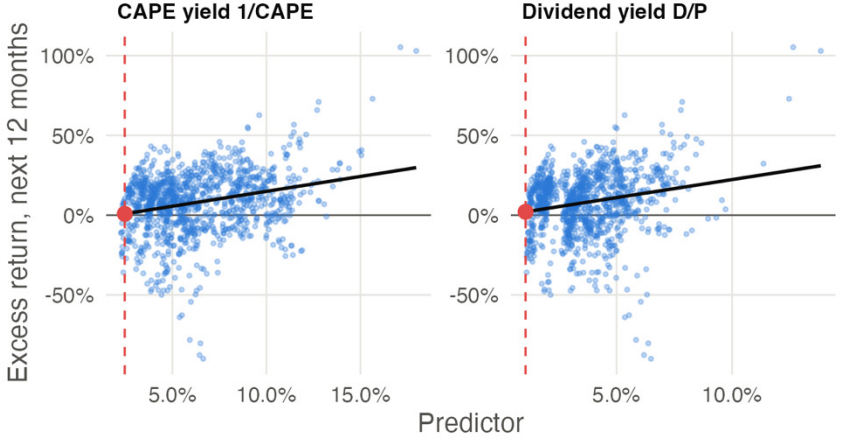
- High valuations predict low returns: CAPE 40.6 is above 99% of months since 1881.
- CAPE and D/P from Shiller, US data since 1926.

2 Predictive regression

$$r^e_{t\to t+12} = a + b \cdot x_t + \varepsilon_{t+12}, \qquad x_t = \mathrm{CAPE}_t^{-1} \text{ or } D_t/P_t$$

$$v^{\mathrm{USD}} = \tfrac{1}{2}\left(\hat{r}^{\,\mathrm{CAPE}} + \hat{r}^{\,D/P}\right) = \tfrac{1}{2}\left(0.85\% + 2.08\%\right) = 1.46\%$$





3 Result

View v (euro)	Equilibrium Π	Slopes: CAPE / D/P	Out-of-sample R^2
1.68%	**4.94%**	**1.87 / 2.26**	**-12.2%**
US equity excess return in euro	equilibrium Π	Newey–West t = 2.96 / 2.04	average (CAPE alone -30.5%)

CAPE and D/P forecasts averaged: 1.46% in USD. In euro: subtract cash-rate gap 0.24% + unhedged carry 0.86% - dollar covariance 0.41% = 1.68%.
Today's valuation lies at the edge of the sample, so the line is extrapolated.

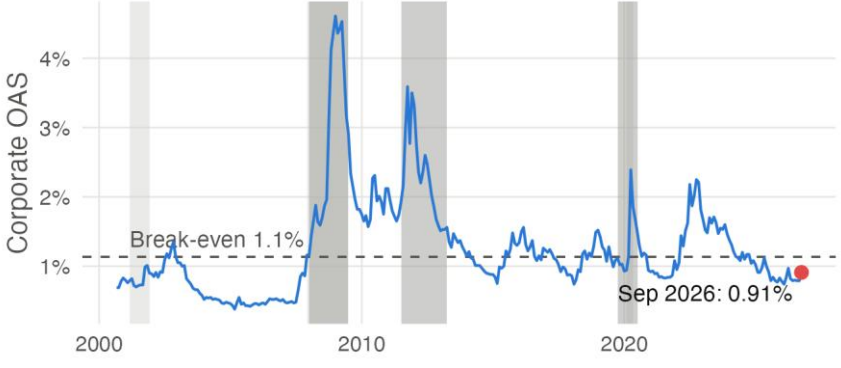
View 4: Tight credit spreads

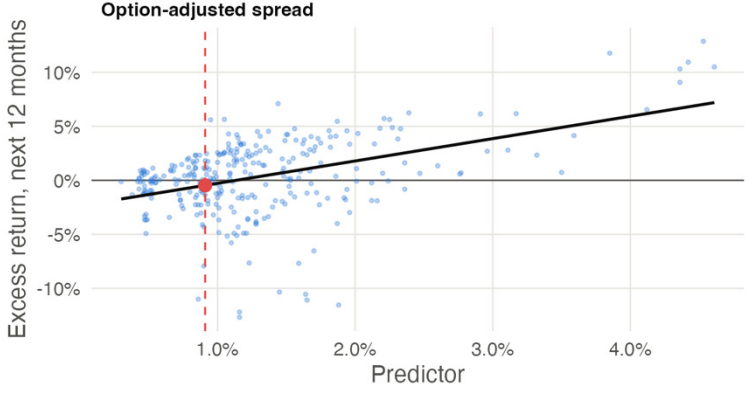
1 Idea

- A wide spread compensates for credit risk.
- A tight spread barely covers expected losses.
- Today's spread 0.91% is below its 1.28% average and the 1.1% break-even.

2 Predictive regression

$$r^{\text{corp}}_{t\to t+12} - r^{\text{govt}}_{t\to t+12} = a + b \cdot \text{OAS}_t + \varepsilon_{t+12}$$





3 Result

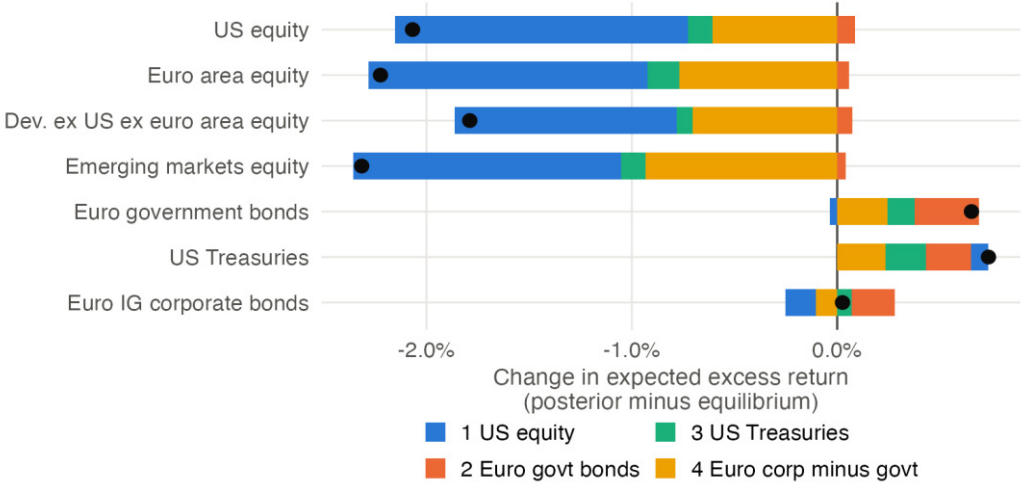
View v (relative)	Equilibrium p'Π	Slope (Newey–West)	Out-of-sample R²
-0.47%	**0.37%**	**2.07 (t = 3.69)**	**4.4%**
corp. minus govt	equilibrium p'Π	R² 19.6%: best in-sample fit	beats the mean (start 2010, the R²_OS is -96.8% with forecasts from 2009)

P row: +1 euro corporates, -1 euro government bonds. Caveat: corporates have a shorter duration (4.3 vs. 6.6 years), so the view also contains a rate bet.

Black-Litterman Model

$$\bar{\mu} = \Pi + \tau\Sigma P' \left(P\tau\Sigma P' + \Omega\right)^{-1} \left(v - P\Pi\right)$$

Prior Π updated by the four views, weighted by their precision (τ = 0.05, full-sample Σ).



Which views move which assets

- **US equity view.** Main driver. Equities are highly correlated, so all four fall, by 1.8 to 2.3 pp. Euro area equity has no view but falls more than US equity because it's more volatile.
- **Credit view.** Credit risk moves with equities, so it accounts for 0.6 to 0.9 pp of that fall. Government bonds are its short leg: euro govt +0.25, US Treasuries +0.24.
- **Bond views.** Raise both government bond classes and, through correlation, euro corporates. The other views offset this, so euro corporates end at +0.03.

% p.a.	US eq.	Euro eq.	Dev. eq.	EM eq.	Euro govt	US Tsy	Euro corp
Equilibrium Π	4.94	5.73	4.52	5.81	0.58	0.06	0.96
Posterior μ̃	**2.87**	**3.51**	**2.73**	**3.50**	**1.24**	**0.80**	**0.98**
Change, pp	-2.07	-2.22	-1.79	-2.32	+0.65	+0.74	+0.03

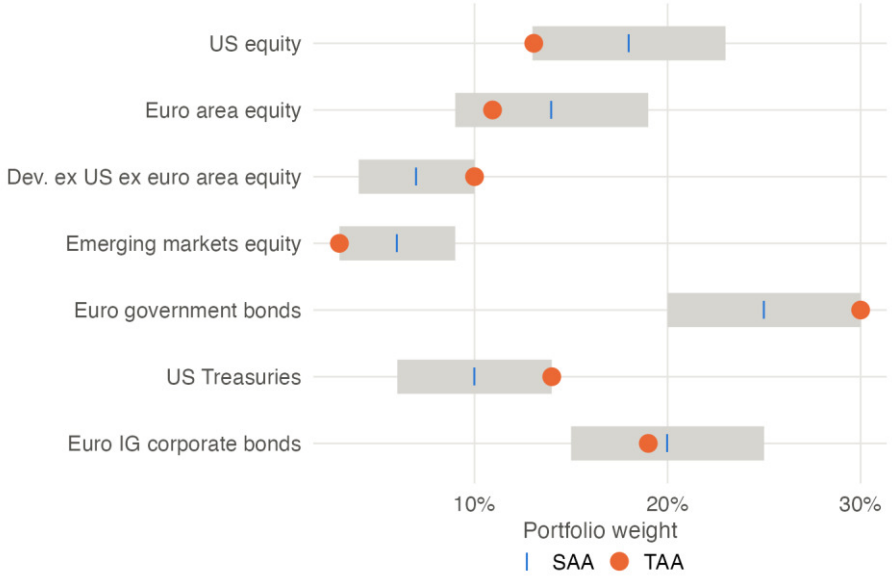
The posterior lies between equilibrium and views (US equity: 2.87%, between 4.94% and 1.68%).

Tactical Asset Allocation

$$\max_{w} \quad w'\bar{\mu} - \frac{A}{2} w'\Sigma w$$

Each asset class stays within 3 to 5 pp of its SAA weight
Total equity 37% to 53%, cash at most 5%, volatility at most 9%.

Asset class	SAA	TAA	Deviation, pp
US equity	18%	**13.1%**	-4.9
Euro area equity	14%	**10.9%**	-3.1
Dev. ex US ex euro eq.	7%	**10.0%**	+3.0
Emerging mkts equity	6%	**3.0%**	-3.0
Euro govt bonds	25%	**30.0%**	+5.0
US Treasuries	10%	**14.0%**	+4.0
Euro IG corp. bonds	20%	**19.0%**	-1.0
Total equity	**45%**	**37.0%**	**-8.0**



Grey bars: tactical bands

TAA against SAA

Exp. excess return	Volatility	Sharpe ratio
1.81%	**6.11%**	**0.30**
SAA 1.99%	SAA 7.23%	SAA 0.28

Why it de-risks

- **Equity falls to its 37% floor:** after the views, equity risk no longer pays.
- **US equity takes the largest cut** (most negative view). Developed equity goes to its upper band (lowest volatility).
- **Euro govt and US Treasuries** hit the upper bands; cash 0%.
- **The 9% limit does not bind** (6.65% with EWMA Σ): A stays at 5.17.

Currency Policy

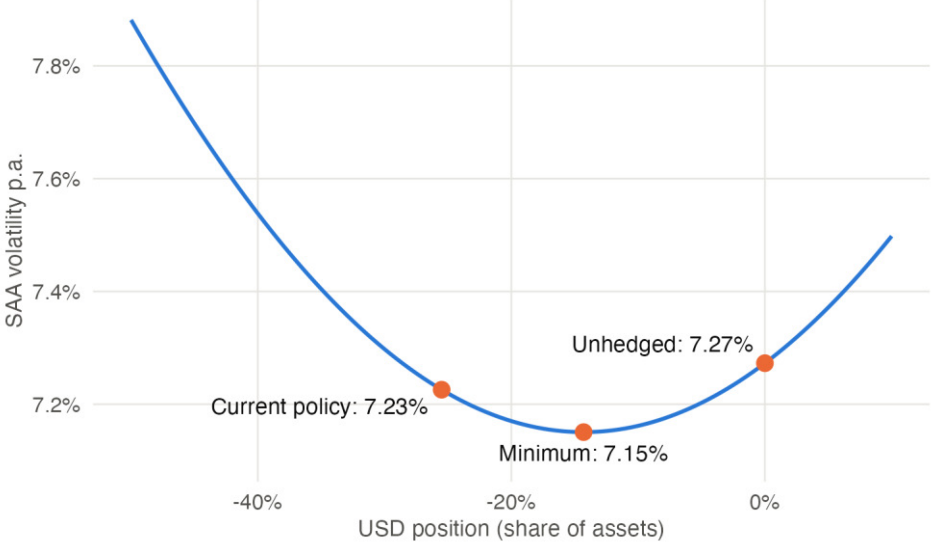
1 **Regress** unhedged excess returns:

$$r_{i,t}^{unh} = \alpha_i + \beta_i x_t + \varepsilon_{i,t}$$

2 **β** is the variance-minimizing hedge ratio.

3 **USD position** from all-unhedged:

$$h^* = -\sum_i w_i \beta_i$$



Asset class	Policy	β (SE)	-wβ
US equity	0.50	0.52 (0.09)	-9.3%
Euro area equity	0.00	-0.25 (0.10)	3.5%
Dev. ex US ex euro eq.	0.50	0.21 (0.08)	-1.4%
Emerging mkts equity	0.50	0.03 (0.11)	-0.2%
Euro govt bonds	0.00	-0.03 (0.03)	0.6%
US Treasuries	1.00	0.90 (0.03)	-9.0%
Euro IG corp. bonds	0.00	-0.08 (0.02)	1.5%
SAA USD pos, h*	**-25.5%**		**-14.3%**

Where the policy fits

US Treasuries (β 0.90) and US equity (β 0.52) are close to the policy.

Where it does not

Developed (β 0.21) and EM equity (β 0.03): a dollar hedge adds risk.

The SAA is over-hedged

h* = -14.3%, against -25.5% under the policy.

Little risk at stake

Volatility: 7.27% unhedged, 7.23% policy, 7.15% at h*.

Recommendation to the Board

	Question	Our recommendation	Why
a	**Is inflation + 2 pp attainable?**	**Only narrowly** Present it to investors as an ambition, but not a promise.	• **4.59% vs. 4.13%:** expected compound return vs. target (2.13% inflation + 2 pp), holding Euribor flat • **About a 58% chance** of reaching it (65% without the glide path) • **Inflation risk:** 55% of the SAA is nominal bonds, which lose when inflation surprises upwards
b	**Do the SAA and the de-risking pace fit?**	**Yes, keep the pace** But do not use euro government bonds alone as the safe asset. Use inflation linked bonds or short duration bonds	• **2008 (sequence risk):** going to 25% equity cuts the loss from 22.3% to 12.7%, and close to retirement there is no time to recover • **2022:** no protection (15.8% vs. 15.9%), as stocks and bonds fell together • **Euro government bonds:** +4.3% in 2008 but -18.6% in 2022 (excess)
c	**Hedge funds for 10 pp of equity?**	**No** Use the existing bonds if more de-risking is needed. The only argument supporting HFs is that they have done better when bonds fell with equities	• **2008:** the Credit Suisse Hedge Fund Index lost 22.4%, as much as the SAA (22.3%) • **Bonds de-risk more:** vol. 6.04% vs. 6.21%, drawdown 24.3% vs. 26.2% • **3.8% p.a. is biased:** survivorship and self-selection. Stale pricing also understates risk; fees, lock-ups and gates add cost
d	**Should we change the currency policy?**	**Yes, a small change** Stop dollar-hedging DM (ex. US, ex EA) and EM equity; keep the rest.	• **Policy fits:** US Treasuries (β 0.90), US equity (β 0.52): close to its 1 and ½ • **Policy does not fit:** developed (β 0.21) and EM equity (β 0.03), where a dollar hedge adds risk • **Result:** USD short falls from 25.5% to 19.0%, volatility is 7.16% (from 7.23%), and it saves 0.11% of carry p.a.